Advanced Machine Learning Techniques for Signal Processing

Deep Learning Approaches for Real-Time Audio and Image Analysis

John Doe

School of Computer and Communication Sciences
Signal Processing Laboratory
Master of Science in Computer Science

Lausanne, June 2026

This page intentionally left blank.

Advanced Machine Learning Techniques for Signal Processing

Deep Learning Approaches for Real-Time Audio and Image Analysis

John Doe

Student No. 123456

Supervisor: Prof. Martin Vetterli
Full Professor, École polytechnique fédérale de Lausanne (EPFL)

Co-supervisor: Dr. Alessandro Cremonesi
Senior Scientist, École polytechnique fédérale de Lausanne (EPFL)

School of Computer and Communication Sciences
Signal Processing Laboratory
Master of Science in Computer Science

Master's Thesis

Lausanne, June 2026

This page intentionally left blank.

Advanced Machine Learning Techniques for Signal Processing

This page intentionally left blank.

Acknowledgements

Writing Guidance

In the *Acknowledgment* section, express your gratitude to those who helped and supported your work. Start by thanking your advisors, mentors, or supervisors who provided guidance and expertise. Mention any colleagues, classmates, or team members who contributed to discussions or offered assistance. You can also acknowledge specific organisations, institutions, or funding sources that supported your research or work. Lastly, include any personal acknowledgments for family or friends who offered encouragement and moral support during the project. Keep this section sincere, concise, and professional.

This page intentionally left blank.

Resumo

Writing Guidance

Na secção *Resumo*, apresente um resumo conciso do seu projeto, destacando os pontos principais. Comece com uma breve declaração do problema ou objetivo, seguido de uma descrição da sua abordagem ou metodologia. Resuma os principais resultados ou conclusões, salientando a sua importância ou implicações. Conclua com uma ou duas frases sobre a contribuição global ou o impacto do seu trabalho. O resumo deve ser claro e conciso, idealmente com 150-250 palavras, para que os leitores compreendam rapidamente o seu trabalho e a sua importância.

Palavras-Chave: Palavra-Chave A, Palavra-Chave B, Palavra-Chave C.

This page intentionally left blank.

Abstract

Writing Guidance

In the *Abstract* section, provide a concise summary of your project, highlighting the key points. Begin with a brief statement of the problem or objective, followed by a description of your approach or methodology. Summarise the main results or findings, emphasising their significance or implications. Conclude with a sentence or two on the overall contribution or impact of your work. Keep the abstract clear and focused, ideally within 150-250 words, to give readers a quick understanding of your research and its importance.

Keywords: Keyword A, Keyword B, Keyword C.

AI Acknowledgement

Writing Guidance

It is good academic practice to briefly state how, why, and when AI was used. Explaining both the actions taken and the reasons behind them encourages critical reflection, deepens understanding of AI's role in learning, and supports the evaluation of its impact on your work's quality and integrity. This page can be removed by setting the option `aiack` to `false` in the document class options.

Example of Use

I acknowledge the use of ChatGPT (https://chatgpt.com) to refine the academic tone and improve the linguistic accuracy of this work, including aspects of grammar, punctuation, and vocabulary.

Description of Use
Prompt 1: In a paragraph, simply describe what LaTeX is.
Output 1: LaTeX is a high-quality typesetting system commonly used for producing scientific and mathematical documents due to its powerful handling of formulas and bibliographies. Unlike word processors, it uses plain text files with markup tags to define the structure and formatting of a document, allowing precise control over layout and appearance. It is especially popular in academia and research for creating theses, journal articles, and presentations.

Contents

Appendices

Annexes

List of Figures

This page intentionally left blank.

List of Tables

This page intentionally left blank.

Glossary

Latex Typesetting system commonly used for the production of scientific and mathematical documents due to its powerful handling of formulas and references. It is widely used for the preparation of documents such as research papers, theses, reports, and articles. *(p. 12)*

Mathematics Mathematics is what mathematicians do. *(p. 12)*

This page intentionally left blank.

Acronyms

GCD Greatest Common Divisor. *(p. 12)*

LCM Least Common Multiple. *(p. 12)*

Symbols

α Angular position of a rotating object, in radians. *(p. 12)*

β Rate of change of angular position, or angular velocity. *(p. 12)*

1

Introduction to the Template Motivation and First Steps

Template: EPFL Thesis
Current Version: 1.0.0
License: LaTeX Project Public License v1.3c
Official Repository: GitHub Repository

Welcome to the *EPFL Thesis* template! Thank you for choosing it for your dissertation, report, or project at the École polytechnique fédérale de Lausanne. This chapter introduces its purpose and helps you get started. See Chapter 2 for a detailed guide, and Chapter 3 for a brief LaTeX tutorial to maximise its use.

1.1 Motivation

This template provides a structured, professional LaTeX foundation for EPFL students writing theses and academic reports. It combines clear file organisation, a clean design aligned with EPFL branding, and flexible class options for language, cover style, and document stage. The goal is to let you focus on your research content rather than document formatting.

1.2 Getting Started

To start using this template, you first need to know how to use LaTeX. For this, please refer to Chapter 3. Once you are familiar with LaTeX, you will need to either install it locally or use an online LaTeX editor.

If you prefer an online editor, Overleaf is a popular choice. To use this template locally:

1. *Clone the repository from GitHub.*
2. *Install a LaTeX distribution (TeX Live, MacTeX, or BasicTeX).*
3. *Edit `Metadata/Metadata.tex` with your thesis details.*
4. *Build with `make all` or `latexmk`.*

If you choose to use a local editor, you must first install LaTeX on your machine. I recommend either TeX Live or MikTeX. After installing LaTeX, select an editor for writing and editing your documents. A helpful overview of editors is available on TeX Stack Exchange. Once LaTeX and your editor are set up, clone the template from GitHub and start writing.

> **Tip**
>
> In this repository, you'll find a Makefile and a Latexmk configuration file, either of which can be used to compile the project. The Makefile supports both `rubber` and `latexmk` as compilers.

1.3 Getting Help

As a newcomer, you may encounter situations where you want to do something in LaTeX or with this template but aren't sure how. When questions arise, you have several options. The TeX Stack Exchange is an active community that can help with nearly any issue. The demo chapters in this document also explain how to use the template. Of course, searching online is always a reliable resource.

1.3.1 Issues and Suggestions

If you encounter a bug or have a suggestion, you can submit them via the *Issues* tab in the GitHub repository. Please be as descriptive as possible when reporting issues.

Pull requests and pushes trigger a GitHub Action that automatically compiles the document. If the compilation fails, the pull request checks will fail. Please keep this in mind when submitting changes.

1.3.2 In-Built Comments, Guidance Texts and Warnings

Within this document template, you may encounter informational text displaying the message "Writing Guidance." These sections are provided solely as guidance to help users understand what content should be included in specific sections. They are not related to the LaTeX code itself within this template.

While navigating through the template, especially the configuration files, you will notice that everything is thoroughly commented. LaTeX can sometimes be difficult to understand without proper documentation for the packages we are working with. With that in mind, advanced modifications are accompanied by detailed comments.

Finally, regarding warnings: please take them into consideration when compiling the document. The template is designed to compile cleanly, so please strive to maintain warning-free builds.

1.4 Important Notices

This template is tailored for EPFL students writing theses, dissertations, and academic reports. Update the metadata in `Metadata/Metadata.tex` with your school, department, supervisors, and thesis details before submitting your work.

If you find this template useful, consider giving the repository a star on GitHub to help others discover it.

2 Comprehensive User Guide Instructions for Using the Template

If you plan to use this template, please read this chapter carefully. It provides all the information you need to effectively use the template, including the mandatory modifications (*e.g.*, title, subtitle, author information) and other configurations that, while not highly recommended, are optional. The template comprises various directories and files, including a total of seven distinct directories and dozens of files. Among these, the most important are `EPFLMain.tex` and `EPFLThesis.cls`. Below, Table 2.1 presents the different directories available, along with their descriptions and a check-mark indicating whether you need to access the directory to make changes. Of course, the check mark indicates that you can make changes to the content, while a hyphen signifies that you should not modify it.

It is crucial to note that the files are organised according to a specific naming convention, which must be **respected** and **maintained**. The naming convention consists of an ascending two-digit numeric value, followed by a hyphen, and then the file name in capital letters. The name should always aim to be a single word. If more than one word is necessary, they should be separated by a hyphen and capitalised.

> **Note**
>
> While Table 2.1 indicates that the *Matter* directory is not modifiable, two files within that directory should be altered when necessary: `05-Glossary.tex` and `06-Acronyms.tex`. Although the names are fairly self-explanatory, these files should contain the glossary and acronyms entries, respectively.

The two files mentioned earlier, `EPFLMain.tex` and `EPFLThesis.cls`, should be

Table 2.1: *Overview of the directory structure in this template.*

Directory	Modifiable	Description
Bibliography	✓	This folder contains the bibliography file used to manage references throughout the document.
Chapters	✓	Individual chapters of the thesis are organised in this directory, making it easy to work on sections separately.
Code	✓	Code examples and relevant scripts are stored here, supporting the content of the thesis.
Configurations	-	All configuration files required for the template, such as layout and style settings, are placed in this directory.
Figures	✓	All figures and images referenced within the document are stored in this folder for easy access and management.
Matter	-	The front matter of the document, including the cover page, copyright statement, and glossary, is assembled in this directory.
Metadata	✓	This folder holds the metadata file, where key document details such as the author, title, and supervisor can be customised.

used with caution. The main file, as the name suggests, is the master file where you will add the necessary chapters to be included in your work. The class file, on the other hand, requires even more caution, and it is not recommended to alter it.

2.1 Template and Class Options

The first thing you need to do is specify the options within the `EPFLMain.tex` file. How do you do that? It's simple. At the top of the file, you will find a `documentclass` command that loads the custom class for this template. In this call, you can pass the options you need. The available options, presented in a key-argument style, are listed in Table 2.2.

Table 2.2: *Class options supported by the template.*

Options	Description
language=OPT *portuguese, english, spanish*	**Language preference selection.** ⇒ *Default: language=english*
chapterstyle=OPT *classic, modern, fancy*	**Selection of a chapter design style.** ⇒ *Default: chapterstyle=classic* *This option modifies the appearance of the chapter, including its title and numbering style. Explore the available styles and apply the one you prefer.*
coverstyle=OPT *classic, bw*	**Choosing a style for the cover.** ⇒ *Default: coverstyle=classic* *classic ⟶ EPFL cover with red gradient background.* *bw ⟶ Black and white cover variant.*

Continued on the next page.

Table 2.2 continued from previous page.

Options	Description
docstage=OPT *final, working*	**Choosing a stage for you document.** *⇒ Default: docstage=final* *final → Assumes this is the final version of the document.* *working → It assumes the document is a work in progress.*
media=OPT *paper, screen*	**Project media type.** *⇒ Default: media=paper* *paper → Blank pages will appear between sections.* *screen → Blank pages will not appear between sections.*
linkcolor=OPT *color*	**Main theme color.** *⇒ Default: linkcolor=red!45!black* *This option requires a valid color name. Refer to the xcolor manual (subsection 4.2) to select a valid color.*
bookprint=OPT *true, false*	**For book printing.** *⇒ Default: bookprint=false* *This option adds a binding margin on odd-numbered pages to allow for printing, as it increases the left margin.*
aiacknowledgement=OPT *true, false*	**AI acknowledgement print.** *⇒ Default: aiacknowledgement=true* *This option adds a section intended for the user to insert their acknowledgement of AI usage.*
listprefix=OPT *true, false*	**Add a prefix to LoF and LoT.** *⇒ Default: listprefix=false* *This options adds the prefix “Figure” or “Table” to both the LoF and LoT when enabled.*

After setting the desired options in the main class, you are all set to personalise the metadata. To learn how, simply refer to Section 2.2.

2.2 Metadata Customisation

While options like language can be passed as arguments to the main class, other options, such as author and title, must be defined manually. Since this template supports a wide range of metadata options, a dedicated file is provided for this purpose. The file at `Metadata/Metadata.tex` lists metadata variables, with comments on whether they are mandatory. Comment out the variables to omit them. Table 2.3 includes all metadata variables, their GET command, and if they are mandatory. The GET command automatically retrieves the information from the stored variable.

Table 2.3: *Metadata variables within the template.*

Variable	Macro Commands	Mandatory
Title	`\GetTitle`	✓

Continued on the next page.

Table 2.3 continued from previous page.

Variable	**Macro Commands**	**Mandatory**
Subtitle	`\GetSubtitle`	✓
University	`\GetUniversity`	✓
School	`\GetSchool`	✓
Department	`\GetDepartment`	✓
Degree	`\GetDegree`	✓
Course	`\GetCourse`	-
Local and date	`\GetDate`	✓
Academic year	`\GetAcademicYear`	✓
Thesis type (*Dissertation, Project or Internship*)	`\GetThesisType`	✓
First author name	`\GetFirstAuthor`	✓
First author identification	`\GetFirstAuthorNumber`	✓
Second author name	`\GetSecondAuthor`	-
Second author identification	`\GetSecondAuthorNumber`	-
Third author name	`\GetThirdAuthor`	-
Third author identification	`\GetThirdAuthorNumber`	-
Supervisor name	`\GetSupervisor`	✓
Supervisor e-mail	`\GetSupervisorMail`	✓
Supervisor title and affiliation	`\GetSupervisorTitle`	✓
Co-supervisor name	`\GetCoSupervisor`	-
Co-supervisor e-mail	`\GetCoSupervisorMail`	-
Co-supervisor title and affiliation	`\GetCoSupervisorTitle`	-
Second co-supervisor name	`\GetSecCoSupervisor`	-
Second co-supervisor e-mail	`\GetSecCoSupervisorMail`	-
Second co-supervisor title and affiliation	`\GetSecCoSupervisorTitle`	-

If you want to add more metadata options, open an issue in the GitHub repository.

2.3 Custom Chapter Insertion

As stated before, to use this template, you need to do three things: set the appropriate options in the document class (see Section 2.1), update the document metadata (see Section 2.2), and create and import your custom chapters. To create and import a custom chapter, follow these steps: i create a TeX file in the Chapters directory that follows the predefined naming convention and $ii)$ include it in the main file using the command `\include{CHAPTER}`. And voilà, your first chapter is ready!

2.4 Custom Commands

Within this template, some custom commands are also available for your use. For example, if you are writing your thesis and want to add a to-do note, you can easily insert a block with the option `todo`, as follows: `\begin{block}[todo]`. This will insert a to-do block with a style similar to Markdown. Other available options are: `tip`, `warning`, and `note`. Below is a visual example for each one.

To-Do

This is a to-do block.

Tip

This is a tip block.

Warning

This is a warning block.

Note

This is a note block.

2.5 EPFL Brand Identity

This template follows the EPFL brand guidelines (corp-id.epfl.ch) wherever a thesis document allows it. This section documents how the guidelines are applied and presents the reusable brand elements that ship with the template.

2.5.1 Corporate Colors

The guidelines state that, when using colors, you must always choose the values of the official palette, and always balance colors with generous white space. The six corporate colors are predefined in the template (see `Configurations/01-Colors.sty`) and can be used anywhere with `\textcolor{epflrouge}{...}` or any other `xcolor` command. Table 2.4 lists the palette with the official print and digital values.

Table 2.4: *The EPFL corporate color palette and its official values.*

	Name	Hex	Pantone / RAL	Macro
	Rouge	`#FF0000`	485 C / 3020	`epflrouge`
	Groseille	`#B51F1F`	3546 CP / 3000	`epflgroseille`
	Taupe	`#413D3A`	405 CP / 8019	`epfltaupe`
	Canard	`#007480`	7714 CP / 5025	`epflcanard`
	Léman	`#00A79F`	3272 CP / 5018	`epflleman`
	Perle	`#CAC7C7`	406 C / 7047	`epflperle`

2.5.2 Logo and Brand Architecture

The EPFL logo always features on the top left side of documents, in Swiss red, primarily on clear backgrounds and images. The logo should remain red as much as possible, even on dark backgrounds; the white version on a red background is reserved for secondary uses. This template honours these rules on the cover, front page, and back page.

The openings of the letters "E" and "F" in the logo come from the proportions of the ends of the arm of the Swiss cross: one sixth wider than high. The same construction is available as the `\epflsquare` element (■), which the guidelines pair with organizational unit references. Unit references must always be separated from the EPFL logo, accompanied by the red square, and set smaller than the logo. Use `\epflunit{...}` to typeset one:

■ School of Computer and Communication Sciences

A branded callout box built on these elements is also available through the `epflbox` environment:

> ■ Branded callout
>
> A white surface with a single Groseille accent and the red square marker, in line with the guideline to balance colors with generous white space.

2.5.3 Typography

EPFL's official corporate font is *Suisse Int'l* (with Arial as the authorised alternative on office materials). As Suisse Int'l is commercially licensed, this template instead adopts Geist, an open-source (SIL OFL 1.1) sans-serif family in the same Swiss typographic tradition. Three members of the family are used, each where pertinent:

Geist Sans The main document font, used for body text, headings, and the cover pages. Light, Regular, **Medium**, **SemiBold**, and **Bold** weights are available through `\geistlight`, `\geistmedium`, and `\geistsemibold`.

Geist Mono Used automatically for all monospaced material: `file paths`, `inline verbatim`, and code listings rendered with `minted`.

Geist Pixel A decorative display family used for the chapter numerals. It comes in five variants, available as font commands:

EPFL 2026	`\geistpixelsquare`
EPFL 2026	`\geistpixelgrid`
EPFL 2026	`\geistpixelcircle`
EPFL 2026	`\geistpixeltriangle`
EPFL 2026	`\geistpixelline`

Warning

Geist Pixel is a display typeface: keep it for headlines, numerals, and accents. Never set body text in it, and never use it where the guidelines require the corporate font.

3

Essential LaTeX Tutorial Fundamentals and Key Concepts

This chapter introduces the LaTeX working environment and the essentials for producing your thesis. LaTeX (pronounced “LAY-tek” or “LAH-tek”) is a tool for creating professional documents using plain text with formatting commands, unlike WYSIWYG editors like Microsoft Word. These files are processed by a TeX engine to generate a polished PDF, allowing you to focus on content while LaTeX handles the layout. While this chapter covers key features, it’s worth learning LaTeX from the start. For a quick introduction, see Overleaf Learn LaTeX series for guidance.

3.1 Citations

We present two distinct approaches for citing entries in the bibliography. The first method involves in-text citations, executed using `\citet{ENTRY}`, while the second method employs `\citep{ENTRY}` for citations within a paragraph. Below is an example that demonstrates both usages. You can cite multiple works in the same citation environment using `\citep{ENTRY1, ENTRY2, ...}`. For citing only the title, use `\citetitle{ENTRY}`, and for the author, use `\citeauthor{ENTRY}`.

> **Tip**
>
> Proper citations are vital in academic writing and ensure credibility, transparency, and knowledge advancement. They are essential for responsible scholarship. Ensure accuracy and appropriateness.

Example: A novel signature scheme is introduced, along with an implementation of the Diffie-Hellman key distribution scheme that accomplishes a public key cryptosystem (Elgamal, 1985). According to Elgamal (1985), a new signature scheme that accomplishes a public key cryptosystem is introduced (...).

3.2 References

Much like citations, it is advisable to employ references in your document for citing crucial elements such as chapters, sections, figures, or tables. To reference these elements, begin by creating a label. This label can be generated using `\label{TEXT}`, and it should be positioned within the element you intend to refer to. Once the element is created, you can utilise `\ref{LABEL}` to generate an in-text reference. **We strongly recommend using** `\autoref{LABEL}`. This command automatically creates a custom link with color corresponding to the type of element being referred to. For instance, a chapter reference will appear like this: Chapter 1, rather than simply Chapter 1.

> **Tip**
>
> Properly referencing elements within the document, such as **chapters, sections, figures, tables, or listings**, is crucial.

3.3 Glossary, Acronyms and Symbols

The document includes a glossary, and an acronym and symbol list, accessible at the beginning of the document. You can create a new entry in either the `Matter/05-Glossary`, `Matter/06-Acronyms` or `Matter/07-Symbols` sections, depending on the type of entry you intend to add. Once the entry is created, you can reference it using `\gls{ENTRY}` for both glossary and symbol entries. For acronym entries, there are two ways to reference them. The first method, `\acrfull{ENTRY}`, should be used the first time the acronym appears in the text as it automatically provides the definition in-text. Subsequently, to refer to the acronym without repeating its meaning, use `\acrshort{ENTRY}`.

Example: Utilising Latex for Mathematics is essential (...). It is advisable to seek both the Greatest Common Divisor (GCD) and Least Common Multiple (LCM) because (...). Subsequently, with the aid of GCD and LCM, we can find both α and β (...).

3.4 Figures

In LaTeX, integrating figures is a straightforward process. To insert them, you should utilise the environment `\begin{figure}`. You can customise the `width` parameter according to your requirements, but it is crucial to select a high-quality figure when inserting it into your documents. It is equally crucial to furnish a well-crafted caption.

If necessary, consider including citations or references to indicate the figure's origin. The caption environment is denoted as `\caption{TEXT}`. To generate a smaller caption for the Table of Figures, be sure to utilise the format `\caption[SMALL_TEXT]{BIG_TEXT}`. By following the aforementioned tips, we can create a figure as demonstrated in Figure 3.1.

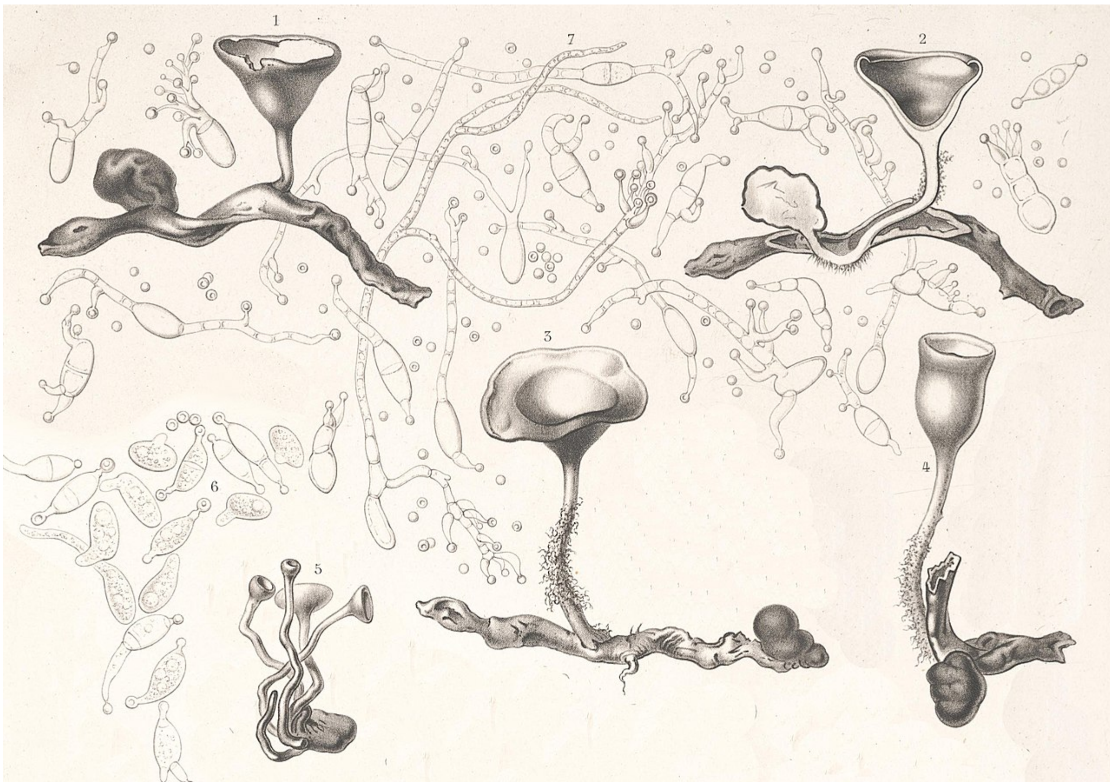
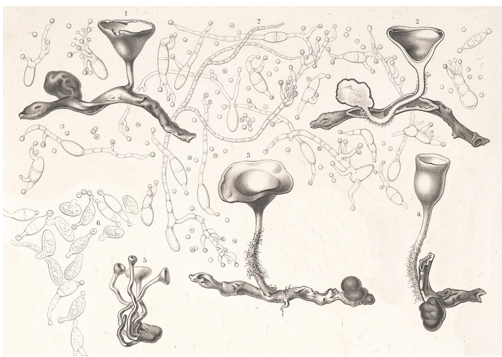
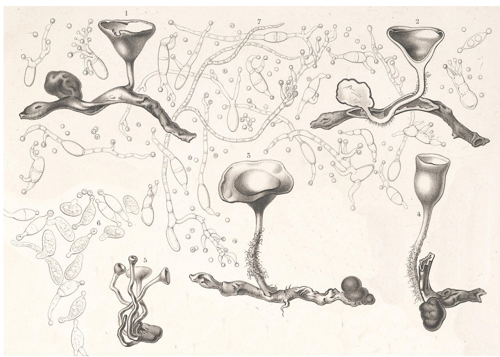


Figure 3.1: *Illustration of the fungus Dumontinia tuberosa by physician, mycologist, and illustrator Charles Tulasne (1816–1884) in the book Selecta Fungorum Carpologia (1861–65). (Name of the original work: Peziza tuberosa parasite on Anemone nemorosa).*

For comparison or for other reasons, you can insert side-by-side figures using both the `\begin{figure}` and `\begin{subfigure}` environments. You can also refer to the sub-figure as Figure 3.2a and Figure 3.2b.



(a) *Caption for figure 1.*



(b) *Caption for figure 2.*

Figure 3.2: *Overall caption of the figure.*

3.5 Tables

Tables are vital for presenting findings effectively. This chapter explores techniques for conveying information through tables using various template environments. Defining tables in LaTeX seems complex, but this template simplifies the process.

Tip

Different table environments must be within a `\begin{table}` environment and use `[!htpb]` float options for better placement. **This advice should be taken into consideration when positioning figures as well**.

3.5.1 Tabular Environment

The conventional `\begin{tabular}` environment enables you to create a simple yet elegant table. Table 3.1 is generated using a centering environment for added emphasis. It also incorporates the `booktab` configuration for a more sophisticated table style.

Table 3.1: *A table showcasing the usage of the tabular environment.*

Header 01	**Header 02**	**Header 03**
Lorem Ipsum	Pharetra Dolor	✓
Amet Consectetuer	Curabitur Aliquet	-
Praesent Mauris	Praesent Libero	✓

3.5.2 Tabularx Environment

Employ the `\begin{tabularx}` package to construct a table featuring automatically expanding multi-columns. To achieve this automatic behaviour for multi-columns, you can use the following environment: `\begin{tabularx}{\textwidth}{lX}`, where `X` is the column that will function as a multi-column. Use `C` to centre the multi-column, and `L` and `R` to align left and right respectively. Table 3.2 showcases the usage of the `\begin{tabularx}` environment.

3.5.3 Longtable Environment

At times, when dealing with exceptionally lengthy tables, it becomes necessary to split them across multiple pages. In LaTeX, this can be achieved using the `\begin{longtable}` environment. This environment is slightly more complex than others, as you need to define the header twice: once for the initial appearance of the table and again for when the table spans additional pages. This repeated header ensures the reader can correctly identify the columns on subsequent pages. Feel free to consult Table 3.3 for a detailed demonstration of how the `longtable` environment works.

Table 3.2: *A table showcasing the usage of the tabularx environment.*

Header 01	**Header 02**
Foo Bar Baz	Quisque cursus, metus vitae pharetra auctor, sem massa mattis sem, at interdum magna augue eget diam.
Ipsum Dolor	Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Curabitur aliquet quam id dui.
Dolor Sit	Phasellus condimentum elementum justo, quis interdum est sagittis ac. Vestibulum non arcu sit amet justo lobortis semper.
Amet Consectetuer	Integer nec odio praesent libero sed cursus ante dapibus diam sed nisi vestibulum non arcu.

Table 3.3: *A table showcasing the usage of the longtable environment.*

Names	**E-Mails**	**Job/Role**
Alice Johnson	alice.johnson@email.com	Project Manager
Bob Thompson	bob.thompson@email.com	Data Analyst
Charlie Davis	charlie.davis@email.com	Marketing Specialist
David Miller	david.miller@email.com	QA Tester
Emily White	emily.white@email.com	Graphic Designer
Frank Martin	frank.martin@email.com	HR Coordinator
Grace Turner	grace.turner@email.com	Financial Analyst
Henry Lee	henry.lee@email.com	System Administrator
Ivy Carter	ivy.carter@email.com	Customer Support
Jack Wilson	jack.wilson@email.com	Frontend Developer
Jane Reed	jane.reed@email.com	UX Designer
Kevin Evans	kevin.evans@email.com	Product Manager
Linda Adams	linda.adams@email.com	Accountant
Mike Hill	mike.hill@email.com	Network Engineer
Nina Garcia	nina.garcia@email.com	Business Analyst
Oliver Smith	oliver.smith@email.com	Sales Representative
Pamela Turner	pamela.turner@email.com	Legal Counsel
Quincy Brown	quincy.brown@email.com	IT Consultant
Rachel Moore	rachel.moore@email.com	Content Writer
Samuel White	samuel.white@email.com	Research Scientist
Amy Harris	amy.harris@email.com	Digital Strategist
Brian Cook	brian.cook@email.com	Operations Manager
Catherine Ross	catherine.ross@email.com	Brand Manager
Daniel Green	daniel.green@email.com	Database Administrator
Emma Taylor	emma.taylor@email.com	Social Media Manager
Felix Carter	felix.carter@email.com	Compliance Officer
Gloria Scott	gloria.scott@email.com	Procurement Specialist
Harold Bennett	harold.bennett@email.com	DevOps Engineer

Continued on the next page.

Table 3.3 continued from previous page.

Names	E-Mails	Job/Role
Isla Cooper	isla.cooper@email.com	User Researcher
James Black	james.black@email.com	Mobile App Developer
Katie Brown	katie.brown@email.com	UI Designer
Leo Perez	leo.perez@email.com	Scrum Master
Megan Clark	megan.clark@email.com	Event Coordinator
Nathan Ward	nathan.ward@email.com	Security Analyst
Olivia Harris	olivia.harris@email.com	Corporate Trainer
Paul King	paul.king@email.com	Territory Manager
Queen Foster	queen.foster@email.com	Paralegal
Rebecca Adams	rebecca.adams@email.com	Copy Editor
Steven Martin	steven.martin@email.com	Robotics Engineer

3.5.4 Complex Tables

Creating intricate tables in LaTeX can be a somewhat challenging task. Therefore, we highly recommend using the Table Generator. With this tool, you can design your table with the desired style and then easily copy and paste it into your document. This approach simplifies the process and helps ensure the accurate representation of complex tables in your LaTeX document. However, it's crucial to keep in mind that a table should be easily comprehensible for the reader and should not be overly complex. **The complexity of a table may impede understanding.** For example, Table 3.4 presents a table with intricate details.

Table 3.4: *A table showcasing the usage of the complex tables.*

Component	Specifications	
	Characteristic	Supported
CPU	Core Count (e.g., 8 Cores)	✓
	Clock Speed (e.g., 3.6 GHz)	✓
	Hyper-Threading	✓
	Integrated Graphics	-
GPU	CUDA Cores (e.g., 5120)	✓
	Base Clock (e.g., 1.5 GHz)	✓
	Ray Tracing Support	✓
	Multi-GPU Support (SLI/CrossFire)	-
Memory	Type (e.g., DDR5, GDDR6)	✓
	Capacity (e.g., 16 GB)	✓
	Memory Bandwidth (e.g., 448 GB/s)	✓
	ECC Support	-
Motherboard Features	PCIe 5.0 Support	✓
	Wi-Fi 6E	✓
	Thunderbolt 4	-

3.6 Lists

Creating lists in LaTeX is straightforward, offering various options to suit your needs. You can generate a bullet list using `\begin{itemize}`, or opt for a numbered list with `\begin{enumerate}`. Below is an example with the `\begin{itemize}` environment.

- List entries start with the `\item` command.
- Individual entries are indicated with a black dot, a so-called bullet.
- The text in the entries may be of any length.

As mentioned earlier, you can generate a numbered list using the `\begin{enumerate}` environment. Here is an example:

1. Items are numbered automatically.
2. The numbers start at 1 with each use of the `enumerate` environment.
3. Another entry in the list.

You can also nest list entries by creating a list inside another list of the same type. Here is an example:

1. First level item
2. First level item
   (a) Second level item
   (b) Second level item
      i. Third level item
      ii. Third level item

Tip

Please note that the labels change automatically regardless of the environment being the same for every list. **This demonstrates that there's no need to worry about changing the environment for something different.**

You can also modify the label of your list to something entirely different that suits your needs. To accomplish this, insert a new `\item` and enclose your desired label in square brackets. For example, `\item[!]` will result in an exclamation point as your new label. Below are some examples of modified labels.

- This is my first point
- Another point I want to make

! A point to exclaim something!

■ Make the point fair and square.

  A blank label?

Finally, you can create a description list. Unlike having a bullet point or a numbered label, a description list enables you to use custom descriptions that suit your list. In the example below, there are three `\item` entries: one without a label, and two with descriptions.

Item 1: This is the first item with a description.
Item 2: Another item with a different description.
An item without a specific label.

3.7 Code Listings

At times, you may want to include source code from your programs and applications within your document. To achieve this, you can use two nested environments: `\begin{listing}` to create a listing with both caption and label, and `\begin{minted}` for code highlighting. Listing 1 provides an example of a source code in C.

Listing 1: *Hello world in C.*

```c
#include <stdio.h>
int main() {
   printf("Hello, World!"); /* printf() outputs the quoted string */
   return 0;
}
```

The code mentioned above was inserted into the document. However, an alternative approach is to input your code from an external file. To do so, you just need to use the command `\inputminted{CODE_LANGUAGE}{FILE}`. Of course, you should place that command inside of the `\begin{listing}` environment. Listing 2 illustrates an example of Haskell source code that has been input from an external file.

Listing 2: *Factorial in Haskell.*

```haskell
-- Factorial function
factorial :: Integer -> Integer
factorial 0 = 1  -- Base case: 0! = 1
factorial n = n * factorial (n - 1)  -- Recursive case

-- Main function to test the factorial
main :: IO ()
main = do
    putStrLn "Enter a number:"
    input <- getLine
    let number = read input :: Integer
    print (factorial number)
```

In some cases, when you simply want to highlight a specific command, it's recommended not to use `listing` or `minted`. Instead, you should utilise the `\verb` command for inline highlighting or the `\begin{verbatim}` environment for longer sections of highlighted code. An example of a lengthy `verbatim` section is provided below, demonstrating how to create a `listing` with an input code:

```
\begin{listing}[!htpb]
    \inputminted{CODE_LANGUAGE}{FILE}
    \caption{TEXT}
    \label{TEXT}
\end{listing}
```

Sometimes it is necessary to display longer code that occupies more than one page. For this purpose, please use the environment `\begin{longlisting}`. This environment will easily break your code into multiple pages for better readability without you worrying about the size of your code. An example is shown below in Listing 3.

Listing 3: *A sample of functions in Lisp.*

```lisp
(defun factorial (n)
  "Calculate the factorial of a number."
  (if (zerop n)
      (* n (factorial (1- n)))))

(defun fibonacci (n)
  "Calculate the nth Fibonacci number."
  (cond ((zerop n) 0)
        ((= n 1) 1)
        (t (+ (fibonacci (1- n)) (fibonacci (- n 2))))))

(defun gcd (a b)
  "Calculate the greatest common divisor of a and b."
  (if (zerop b)
      a
      (gcd b (mod a b))))

(defun primes-up-to (limit)
  "Return a list of all prime numbers up to LIMIT."
  (let ((primes '()))
    (loop for i from 2 to limit
          unless (some (lambda (p) (zerop (mod i p))) primes)
          do (push i primes))
    (nreverse primes)))

(defun example-function (x)
  "An example function to demonstrate Lisp capabilities."
  (let ((result (list (factorial x)
                      (fibonacci x)
                      (gcd x 10)
                      (primes-up-to x))))
    (format t "Factorial of ~A: ~A~%" x (factorial x))
    (format t "Fibonacci of ~A: ~A~%" x (fibonacci x))
```

```
    (format t "GCD of ~A and 10: ~A~%" x (gcd x 10))
    (format t "Primes up to ~A: ~A~%" x (primes-up-to x))
    result))

(example-function 10)
```

3.8 Equations

When writing equations and other mathematical expressions, LaTeX is a powerful and versatile tool. You can enter a formula in inline mode using the environment `\(FORMULA\)` or use `\begin{equation}` to display it in "math mode" with numbering. If you prefer not to display the equation number, you can use the environment `\[FORMULA\]`.

Example: In physics, the mass-energy equivalence is expressed by the equation $E = mc^2$, discovered in 1905 by Albert Einstein. In natural units ($c = 1$), the formula (3.1) expresses the identity:

$$E = m \tag{3.1}$$

Example: Below is a equation – *without numbering* – for the regularised loss function in supervised learning, combining the average prediction loss over the training dataset and an L_2 regularisation term to prevent overfitting:

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^{N} \ell(y_i, f(\mathbf{x}_i; \boldsymbol{\theta})) + \lambda \|\boldsymbol{\theta}\|_2^2$$

Equations can be a bit challenging to create, so we advise using an online editor, like the LaTeX Equation Editor. Simply build your formulas there and copy and paste them into your document, either inline or in a math block, as shown above.

3.9 Footnotes

Sometimes it is important to present information that is not central to the main text in a footnote. In LaTeX this can be easily achieved using the command `\footnote{TEXT}`. The text will appear at the bottom of the page[1].

If you want to use footnotes within tables, it is best to reconsider, as LaTeX does not provide an easy way to handle them. Instead, you can place a "*" wherever you want the footnote reference to appear. Then, below the table **but before ending the table environment**, place the "*" along with the footnote text. This will create a similar footnote, but it will appear below the table rather than at the bottom of the page.

[1] This is a simple footnote.

This page intentionally left blank.

Bibliography

Elgamal, T. (July 1985). “A public key cryptosystem and a signature scheme based on discrete logarithms”. In: *IEEE Transactions on Information Theory* 31.4, pp. 469–472. ISSN: 1557-9654. DOI: `10.1109/TIT.1985.1057074`. URL: `https://ieeexplore.ieee.org/document/1057074`.

This page intentionally left blank.

Appendices

A
Showcasing the First Appendix

Writing Guidance

Appendices contain supplementary material **created by the author** that enhances the reader's understanding of the dissertation while not being essential for following the primary narrative. These sections often include detailed tables, figures, complex calculations, or materials like survey questions and interview transcripts produced in the course of the research. The appendices allow readers to explore the research in greater detail, offering a deeper insight into methods and findings without interrupting the main body of work.

B
Showcasing the Second Appendix

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

This page intentionally left blank.

Annexes

L
Showcasing the First Annex

Writing Guidance

Annexes are supplementary sections in a dissertation that provide additional information or external documents not essential to the main arguments but that support or complement the research. Unlike appendices, **annexes generally contain material that was not developed by the author**, such as reports, legal documents, or published datasets from external sources. This information is placed separately to keep the main content concise, allowing readers access to relevant external references without disrupting the dissertation's flow.

This page intentionally left blank.